A Structural Generative Control Framework: Behavior Mode Generation under Structural Constraints

Abstract

This paper proposes a Structural Generative Control Framework (SGCF), a novel paradigm in which control is formulated as behavior generation under structural constraints rather than direct action computation.

Unlike traditional control approaches that map system state to control actions, SGCF introduces an intermediate behavioral layer defined by a finite set of behavior modes. These modes emerge as stable solutions of an underlying structure composed of driving forces, relational constraints, and system limitations.

We show that in critical scenarios such as goal-obstacle conflicts, instability, and rapid mode switching, SGCF maintains system feasibility where conventional control methods fail. The framework enables continuous transitions between behavior modes, avoiding discontinuities and collapse.

SGCF provides a unified formulation bridging structure, behavior, and control, and represents a potential new paradigm for organizing control systems.

1. Introduction

Conventional control methods such as PID, MPC, and RL compute actions directly from system states. However, they often fail in scenarios involving conflicts, instability, or discontinuities.

SGCF introduces a behavioral layer that organizes system behavior into structured modes. Instead of computing actions directly, control is achieved by selecting and generating behavior modes under structural constraints.

Conventional control computes actions, while SGCF organizes behaviors.

2. Framework Definition

Core formulation:

$B = G(S, I)$

Expanded:
S → M(S) → Mode → B → u

Structure (S): goals, obstacles, relations, constraints
Mode Space (M): behavior modes generated from structure
Mode: current behavioral state

3. Behavior Mode Dictionary

Core modes:
- Approach
- Avoid
- Stabilize
- Bypass
- Idle
- Recovery

Behavior modes are not predefined rules but stable solutions of structure.

4. Mode Transition Law

Mode = argmin(C + R - S)

C: Conflict
R: Risk
S: Stability

Principles:
- Sustainability
- Minimum conflict
- Stability priority

5. Structural Mode Generation

Modes emerge as stable solutions of structure.

Examples:
- Goal dominant → Approach
- Obstacle dominant → Avoid
- Conflict → Bypass
- Instability → Stabilize

6. Validation

Scenario 1: Conflict → SGCF bypasses obstacle
Scenario 2: Instability → SGCF stabilizes system
Scenario 3: Switching → SGCF ensures smooth transitions

7. Discussion

SGCF is not optimization-based and not rule-based.
It is a structural constraint system that organizes behavior.

8. Conclusion

Control is not the computation of actions, but the generation of behaviors under structural constraints.